Lecture 2

Quotients, products, and the symmetric group

Objectives. *Lecture 1 showed that the cosets G/H form a G-set but in general not a group. This lecture finds the subgroups for which they do, the normal ones, and with them quotient groups, the isomorphism theorems and products. It then finds the conjugacy classes of S_n, indexed by partitions, and the sign.*

By the end of this lecture you will be able to:

- Decide whether a subgroup is normal, and identify a quotient by the first isomorphism theorem.
- List the subgroups between N and G from those of G/N.
- Recognise a group as a direct or semidirect product of a normal subgroup and a complement.
- Find the cycle type, class size, centraliser order and sign of a permutation, and write it as a product of transpositions.

2.1 Normal subgroups and quotients

In S_3 the rule $xH \cdot yH = xyH$ failed for $H = \langle (1\,2) \rangle$ (Example 1.9), because conjugation moves H: $(1\,3)(1\,2)(1\,3)^{-1} = (2\,3) \notin H$.

Definition 2.1 (Normal subgroup)**.** A subgroup $N \leq G$ is *normal*, written $N \trianglelefteq G$, if $gNg^{-1} = N$ for every $g \in G$; that is, if $N_G(N) = G$ (Corollary 1.17).

It is enough that $gNg^{-1} \subseteq N$ for every g: applied to g^{-1} this gives $g^{-1}Ng \subseteq N$, that is, $N \subseteq gNg^{-1}$. Multiplying $gNg^{-1} = N$ on the right by g, normality says $gN = Ng$ for every g. Every subgroup of an abelian group is normal, and so is $Z(G)$. The kernel of a homomorphism φ is normal: if $\varphi(k) = e$, then

$$\varphi(gkg^{-1}) = \varphi(g)\varphi(g)^{-1} = e.$$

Proposition 2.2 (Quotient groups)**.** *Let $H \leq G$.*

(i) The rule $xH \cdot yH = xyH$ is well defined on G/H if and only if $H \trianglelefteq G$. Then G/H is a group, the quotient *of G by H, with identity H and $(xH)^{-1} = x^{-1}H$; and the* quotient map *$q\colon G \to G/H$, $g \mapsto gH$, is a surjective homomorphism with kernel H.*

(ii) $H \trianglelefteq G$ if and only if H is a union of conjugacy classes of G.

(iii) If $[G : H] = 2$, then $H \trianglelefteq G$.

Proof. Recall that $xH = yH$ exactly when $x^{-1}y \in H$.

(i) Let $H \trianglelefteq G$, and let $x' = xh$ and $y' = yk$ with $h, k \in H$. Then

$$x'y' = xy\,(y^{-1}hy)\,k$$

with $y^{-1}hy \in H$, so $x'y'H = xyH$.

Conversely, suppose the rule is well defined, and let $h \in H$ and $g \in G$. As $hH = eH$,

$$hg^{-1}H = (hH)(g^{-1}H) = (eH)(g^{-1}H) = g^{-1}H,$$

so $(g^{-1})^{-1}hg^{-1} = ghg^{-1} \in H$. Hence $gHg^{-1} \subseteq H$ for every g.

The group axioms in G/H follow from those in G, the rule says $q(xy) = q(x)q(y)$, and $q(g) = H$ exactly when $g \in H$.

(ii) $gHg^{-1} \subseteq H$ for all g says that with each h, H contains its class $[h]$.

(iii) The two left cosets partition G, so they are H and $G \setminus H$; as $xH \mapsto Hx^{-1}$ is a bijection, so are the two right cosets. So $gH = Hg$ for every g. ∎

So the normal subgroups are exactly the kernels: N is the kernel of $q\colon G \to G/N$. In S_3, $A_3 = \langle (1\,2\,3) \rangle$ has index 2, so S_3/A_3 has order 2. Lecture 3 uses index-two normality and the union-of-classes criterion to rule out order 6 in A_4.

2.2 The isomorphism and correspondence theorems

Theorem 2.3 (Isomorphism theorems). *Let $\varphi\colon G \to K$ be a homomorphism, $N \trianglelefteq G$, and $q\colon G \to G/N$ the quotient map.*

(i) (Factorisation.) If $N \leq \ker\varphi$, there is exactly one homomorphism $\bar{\varphi}\colon G/N \to K$ with $\bar{\varphi}(gN) = \varphi(g)$, that is, $\varphi = \bar{\varphi} \circ q$.

(ii) (First isomorphism theorem.) $G/\ker\varphi \cong \operatorname{im}\varphi$, by $g\ker\varphi \mapsto \varphi(g)$.

(iii) (Correspondence.) $H \mapsto q(H) = H/N$ is a bijection from the subgroups H of G that contain N to the subgroups of G/N, with inverse $\bar{H} \mapsto q^{-1}(\bar{H})$. It preserves inclusion and index, $[G : H] = [G/N : H/N]$, and $H \trianglelefteq G$ if and only if $H/N \trianglelefteq G/N$.

Proof. (i) If $gN = g'N$, write $g' = gn$ with $n \in N \leq \ker\varphi$. Then $\varphi(g') = \varphi(g)$, so $\bar{\varphi}$ is well defined. Multiplicativity follows from φ; surjectivity of q forces uniqueness.

(ii) Take $N = \ker\varphi$, normal by § 2.1. The map $\bar{\varphi}$ from factorisation is onto $\operatorname{im}\varphi$. Its kernel is trivial, since $\bar{\varphi}(gN) = e$ exactly when $g \in N$, so it is injective (Lecture 1).

(iii) *The bijection.* Let $N \leq H \leq G$. If $q(g) = q(h)$ with $h \in H$, then $g \in hN \subseteq H$; so

$$q^{-1}(q(H)) = H.$$

For $\bar{H} \leq G/N$, $q^{-1}(\bar{H})$ is a subgroup containing $N = q^{-1}(\{N\})$, and $q(q^{-1}(\bar{H})) = \bar{H}$, as q is onto. So the two maps are inverse bijections, and both preserve inclusion.

The index. By the bijection,

$$gH = g'H \iff g^{-1}g' \in H \qquad \iff q(g)^{-1}q(g') \in q(H),$$

so $gH \mapsto q(g)\,q(H)$ is a bijection $G/H \to (G/N)/(H/N)$.

Normality. For $g \in G$, gHg^{-1} is a subgroup containing $gNg^{-1} = N$, and

$$q(gHg^{-1}) = q(g)\,q(H)\,q(g)^{-1}.$$

By the bijection, $gHg^{-1} = H$ exactly when $q(g)\,q(H)\,q(g)^{-1} = q(H)$; and $q(g)$ runs over all of G/N. ∎

Lecture 6 lifts a representation by composing with q. Factorisation gives the converse when N lies in the kernel.

2.3 Direct and semidirect products

The *direct product* $G \times H$ consists of pairs, with $(g,h)(g',h') = (gg', hh')$.

For finite G, H, $(g,h)^k = e$ exactly when both $o(g)$ and $o(h)$ divide k. Thus $o((g,h)) = \operatorname{lcm}(o(g), o(h))$. In $C_m \times C_n$, a pair of generators has the largest possible order, $\operatorname{lcm}(m,n)$. The product is therefore cyclic exactly when $\gcd(m,n) = 1$.

Example 2.4 (Two products of cyclic groups)**.** Consequently

$$C_2 \times C_3 \cong C_6,$$

whereas $C_2 \times C_2$ is not cyclic.

When is a group a product of two of its subgroups?

Theorem 2.5 (Semidirect products)**.** *Let $N \trianglelefteq G$ and $H \leq G$ with $N \cap H = \mathbf{1}$.*

(i) $NH = \{nh : n \in N,\ h \in H\}$ *is a subgroup, and each of its elements is nh for exactly one pair (n,h); so $|NH| = |N|\,|H|$.*

(ii) In NH, with $hn'h^{-1} \in N$,

$$(nh)(n'h') = (n \cdot hn'h^{-1})(hh').$$

(iii) If moreover $H \trianglelefteq G$, then $nh = hn$ for all $n \in N$, $h \in H$, and $(n,h) \mapsto nh$ is an isomorphism $N \times H \to NH$.

If $NH = G$, as when G is finite and $|N|\,|H| = |G|$, then G is the (internal) semidirect product *$N \rtimes H$; H is a* complement *to N. If both factors are normal, $G \cong N \times H$.*

Proof. (i) Insert $h^{-1}h$:

$$nh\,n'h' = n\,(hn'h^{-1})\,hh',$$

Since $hn'h^{-1} \in N$, this proves closure under products. Also $e \in NH$, and

$$(nh)^{-1} = h^{-1}n^{-1} = (h^{-1}n^{-1}h)\,h^{-1} \in NH.$$

If $nh = n'h'$, then $n'^{-1}n = h'h^{-1}$ lies in $N \cap H = \mathbf{1}$, so $n = n'$ and $h = h'$.

(ii) The same insertion gives the stated multiplication rule.

(iii) Group the element $nhn^{-1}h^{-1}$ in two ways:

$$nhn^{-1}h^{-1} = n\,(hn^{-1}h^{-1}) \in N,$$
$$nhn^{-1}h^{-1} = (nhn^{-1})\,h^{-1} \in H.$$

So the commutator is e, and $nh = hn$. The multiplication formula becomes $(nh)(n'h') = (nn')(hh')$, so the unique factorisation gives an isomorphism. ∎

The multiplication rule uses only N, H and conjugation, $n' \mapsto hn'h^{-1}$. Lecture 3 uses the counting form, $|N|\,|H| = |G|$, for groups of order pq.

Example 2.6 (The Klein four-group)**.** The group

$$V_4 = \{e,\ (1\,2)(3\,4),\ (1\,3)(2\,4),\ (1\,4)(2\,3)\}$$

is abelian, so its subgroups $\langle(1\,2)(3\,4)\rangle$ and $\langle(1\,3)(2\,4)\rangle$ are normal. They meet in $\mathbf{1}$, and $2 \cdot 2 = 4$; so $V_4 \cong C_2 \times C_2$.

2.4 S_n: cycles, conjugacy and z_μ

The cycles of a permutation (Lecture 1) are the orbits of an action.

Proposition 2.7 (Cycle decomposition)**.** *Let $\sigma \in S_n$ and $a \in \{1, \dots, n\}$, and let $k \geq 1$ be least with $\sigma^k(a) = a$. The orbit of a under $\langle\sigma\rangle$ is $\{a, \sigma(a), \dots, \sigma^{k-1}(a)\}$, with k distinct elements. So σ is the product of the cycles $(a\ \sigma(a)\ \dots\ \sigma^{k-1}(a))$, one for each orbit, and these commute.*

Proof. Such a k exists, as $\sigma^{o(\sigma)} = e$. If $\sigma^i(a) = \sigma^j(a)$ with $0 \leq i < j < k$, then $\sigma^{j-i}(a) = a$, against the choice of k; and $\sigma^m(a)$ depends only on m modulo k. So the orbit is as stated, and σ acts on it as the cycle.

Every point lies in one orbit, and only its own cycle moves it, so σ and the product of the cycles agree at every point. Cycles that move disjoint sets commute, for the same reason. ∎

The decomposition is unique up to the order of the cycles and the starting point of each, since the orbits are determined by σ.

Definition 2.8 (Partitions and cycle type)**.** A *partition* $\lambda \vdash n$ is a sequence $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_\ell > 0$ of integers with sum n. Its *multiplicity* $m_i(\lambda)$ is the number of parts equal to i; we write $(2, 1^2) = (2, 1, 1)$; and $p(n)$ is the number of partitions of n.
The *cycle type* of $\sigma \in S_n$ is the partition μ of its cycle lengths, fixed points counted as cycles of length 1. For a partition μ,

$$z_\mu = \prod_{i \geq 1} i^{m_i(\mu)}\, m_i(\mu)!\,.$$

A cycle of length i has order i, and the cycles of σ commute and move disjoint sets, so $\sigma^k = e$ exactly when every part of μ divides k:

$$o(\sigma) = \operatorname{lcm}(\mu_1, \dots, \mu_\ell).$$

Theorem 2.9 (Classes of S_n)**.**

(i) For $\tau \in S_n$,

$$\tau\,(a_1\,a_2\,\dots\,a_k)\,\tau^{-1} = (\tau(a_1)\,\tau(a_2)\,\dots\,\tau(a_k)).$$

(ii) $\sigma, \sigma' \in S_n$ are conjugate if and only if they have the same cycle type. So the classes of S_n correspond to the partitions of n, and there are $p(n)$ of them.

(iii) The class of type μ has $n!/z_\mu$ elements, and $\left|C_{S_n}(\sigma)\right| = z_\mu$ for σ of type μ.

Proof. (i) Let $\sigma = (a_1\ \dots\ a_k)$, indices read modulo k. Then

$$\tau\sigma\tau^{-1}\colon\ \tau(a_i) \longmapsto \tau(a_{i+1}),$$

and $\tau\sigma\tau^{-1}$ fixes $\tau(b)$ for every other b, since σ fixes b.

(ii) Conjugation relabels entries and keeps lengths, by the cycle formula. Conversely, if σ and σ' have the same type, write their cycles, fixed points included, σ' under σ with equal lengths aligned:

$$\sigma = (a_1\ \dots\ a_k)(b_1\ \dots\ b_l)\cdots,$$
$$\sigma' = (a'_1\ \dots\ a'_k)(b'_1\ \dots\ b'_l)\cdots.$$

The map τ sending each entry to the one below it is a permutation, and $\tau\sigma\tau^{-1} = \sigma'$ by that formula.

(iii) Fill the pattern of μ, one cycle of each part's length, with $1, \dots, n$ in each of the $n!$ ways. Each filling gives a permutation of type μ, and each permutation of type μ arises.

By the uniqueness of the decomposition, the fillings that give the same σ differ by rotating cycles (i ways for each of length i) and permuting cycles of equal length ($m_i(\mu)!$ ways): z_μ distinct fillings. So, with Theorem 1.15,

$$|[\sigma]| = \frac{n!}{z_\mu},$$
$$\left|C_{S_n}(\sigma)\right| = \frac{n!}{|[\sigma]|} = z_\mu. \quad \blacksquare$$

The inverse of a cycle is the reversed cycle,

$$(a_1\, a_2\ \dots\ a_k)^{-1} = (a_k\ \dots\ a_2\, a_1),$$

so σ^{-1} has the type of σ: every permutation is conjugate to its inverse. Lecture 10 finds a conjugating element that fixes n.

2.5 Sign, generators and A_n

The *permutation matrix* of $\sigma \in S_n$ is the $n \times n$ matrix $D(\sigma)$ with entries $D(\sigma)_{ij} = \delta_{i,\sigma(j)}$: its jth column is $e_{\sigma(j)}$. Then for each j

$$D(\sigma)D(\tau)e_j = D(\sigma)e_{\tau(j)} = e_{\sigma\tau(j)} = D(\sigma\tau)e_j,$$

so $D(\sigma\tau) = D(\sigma)D(\tau)$.

The *sign* of σ is $\operatorname{sgn}\sigma = \det D(\sigma)$. A *transposition* is a 2-cycle $(i\, j)$. A permutation is *even* for sign $+1$ and *odd* for sign -1.

Theorem 2.10 (Sign).

(i) $\operatorname{sgn}\colon S_n \to \{\pm 1\}$ *is a homomorphism, and the sign of a transposition is* -1.

(ii) *A k-cycle is a product of $k-1$ transpositions,*

$$(a_1\, a_2\ \dots\ a_k) = (a_1\, a_k)(a_1\, a_{k-1})\cdots(a_1\, a_2),$$

so $\operatorname{sgn}\sigma = (-1)^{n-c(\sigma)}$, *where $c(\sigma)$ is the number of cycles of σ, fixed points included.*

(iii) *For $n \geq 2$, the* alternating group $A_n = \ker\operatorname{sgn}$ *is a normal subgroup of index 2, with $n!/2$ elements.*

Proof. (i) The determinant is multiplicative, so sgn is a homomorphism to $\mathbb{C}^\times$. The columns of $D(\sigma)$ are distinct standard basis vectors, so $D(\sigma)^{\mathsf{T}}D(\sigma) = I$ and $\operatorname{sgn}\sigma = \pm 1$.

$D((i\, j))$ is the identity matrix with columns i and j exchanged, and exchanging two columns changes the sign of a determinant.

(ii) Read the product right to left. Each a_i is fixed by every factor that does not contain it, so

$$a_1 \mapsto a_2, \qquad a_i \mapsto a_1 \mapsto a_{i+1} \quad (2 \le i < k), \qquad a_k \mapsto a_1.$$

By multiplicativity a k-cycle has sign $(-1)^{k-1}$. For type μ,

$$\operatorname{sgn}\sigma = (-1)^{\sum_j (\mu_j - 1)} = (-1)^{n - c(\sigma)}.$$

(iii) For $n \ge 2$, $\operatorname{sgn}(1\,2) = -1$, so sgn is onto $\{\pm 1\}$, and $S_n/A_n \cong \{\pm 1\}$ by Theorem 2.3. ∎

The sign is constant on classes; for $n = 1$ it is trivial, so (iii) needs $n \ge 2$.

Proposition 2.11 (Generators). *S_n is generated by its transpositions, and also by $s_1, \dots, s_{n-1}$, where $s_i = (i\ i{+}1)$.*

Proof. The first claim follows from Proposition 2.7 and Theorem 2.10(ii). For the second it is enough to reach each $(i\,j)$ with $i < j$. Let

$$w = s_{j-1} s_{j-2} \cdots s_{i+1}.$$

Right to left, its factors carry $i+1$ to $i+2, \dots, j$, so $w(i+1) = j$, while $w(i) = i$. By Theorem 2.9(i),

$$w s_i w^{-1} = (w(i)\ w(i{+}1)) = (i\,j).$$ ∎

The s_i are the *Coxeter generators*. Lecture 11 and Lecture 12 describe a representation of S_n by the matrices of $s_1, \dots, s_{n-1}$ alone, which determine it by Proposition 2.11.

2.6 Worked example

Find the classes, normal subgroups and product structure of S_4.

Cycle types (Theorem 2.9) and signs (Theorem 2.10) give:

type μ	representative	z_μ	class size $24/z_\mu$	order	sign
(1^4)	e	24	1	1	+1
$(2, 1^2)$	$(1\,2)$	4	6	2	−1
(2^2)	$(1\,2)(3\,4)$	8	3	2	+1
$(3, 1)$	$(1\,2\,3)$	3	8	3	+1
(4)	$(1\,2\,3\,4)$	4	6	4	−1

Example 2.12 (Guard: class size is not element order). The 3 elements of type (2^2) and the 6 of type $(2, 1^2)$ all have order 2: the order is the least common multiple of the parts.

Normal subgroups. A normal subgroup is a union of classes containing e (Proposition 2.2), of order dividing 24. Adding class sizes to 1 gives only

$$\{1, 4, 12, 24\},$$

all realised by subgroups. V_4 is e with the class (2^2), closed as $(1\,2)(3\,4) \cdot (1\,3)(2\,4) = (1\,4)(2\,3)$; A_4 is the even types. So the normal subgroups of S_4 are $\mathbf{1}$, V_4, A_4, S_4.

The quotient. S_4 permutes the three ways of splitting $\{1,2,3,4\}$ into two pairs,

$$P_1 = \{12\,|\,34\}, \qquad P_2 = \{13\,|\,24\}, \qquad P_3 = \{14\,|\,23\}.$$

Numbering P_k as k, this is a homomorphism $\pi\colon S_4 \to S_3$ (Proposition 1.7). For instance,

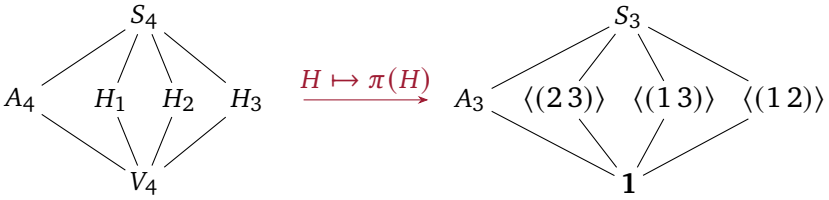
$$\pi((1\,2)) = (2\,3), \qquad \pi((1\,2\,3)) = (1\,3\,2).$$

Each element of V_4 fixes every P_k, so $\ker\pi$ is a normal subgroup containing V_4 but not $(1\,2\,3)$: it is V_4. So π has $24/4 = 6$ values, is onto, and $S_4/V_4 \cong S_3$ (Theorem 2.3).

The correspondence. By Theorem 2.3(iii) the subgroups of S_4 containing V_4 correspond to those of S_3: $\mathbf{1}$, the three $\langle(i\,j)\rangle$, A_3 and S_3 (Corollary 1.12). The preimage of $\langle(2\,3)\rangle$, the stabiliser of 1, is the stabiliser H_1 of P_1, of order 8:

$$H_1 = V_4 \cup \{(1\,2),\ (3\,4),\ (1\,3\,2\,4),\ (1\,4\,2\,3)\}.$$

Likewise H_2, H_3; and A_3 pulls back to A_4, the only subgroup of index 2:



Lecture 3 meets H_1, H_2, H_3 again as the Sylow 2-subgroups of S_4.

The product. Let $S_3 \leq S_4$ be the stabiliser of 4. Each $v \neq e$ in V_4 moves 4, so $V_4 \cap S_3 = \mathbf{1}$; and $|V_4|\,|S_3| = 24$. So $S_4 = V_4 \rtimes S_3$ (Theorem 2.5). It is not a direct product: S_3 holds three of the six transpositions, so it is not normal.

Example 2.13 (Guard: normality is not transitive). $\langle(1\,2)(3\,4)\rangle \trianglelefteq V_4$, as V_4 is abelian, and $V_4 \trianglelefteq A_4$. But by Theorem 2.9(i),

$$(1\,2\,3)\,(1\,2)(3\,4)\,(1\,2\,3)^{-1} = (2\,3)(1\,4),$$

so $\langle(1\,2)(3\,4)\rangle$ is not normal in A_4. In Lecture 4's composition series $S_4 \trianglerighteq A_4 \trianglerighteq V_4 \trianglerighteq \langle(1\,2)(3\,4)\rangle \trianglerighteq \mathbf{1}$, each term need only be normal in the one before.

Self-check. Show that $D_n = C_n \rtimes C_2$, with $C_n = \langle r\rangle$ and $C_2 = \langle s\rangle$, not a direct product for $n \geq 3$. *Answer:* $\langle r\rangle$ has index 2, so it is normal; $\langle r\rangle \cap \langle s\rangle = \mathbf{1}$ by determinants; $n \cdot 2 = |D_n|$; and s acts by $srs^{-1} = r^{-1}$. As $rsr^{-1} = r^2 s \neq s$ for $n \geq 3$, $\langle s\rangle$ is not normal.

Summary.

- *The main results:* G/N is a group exactly when N is normal (Proposition 2.2); $G/\ker\varphi \cong \operatorname{im}\varphi$, and the subgroups of G/N correspond to those of G above N (Theorem 2.3); $N \cap H = \mathbf{1}$ with N normal gives $NH = N \rtimes H$ (Theorem 2.5); the classes of S_n are the cycle types, with centralisers of order z_μ (Theorem 2.9).
- *Handed on:* quotients and the isomorphism theorems; the sign and A_n; the generators s_i; $S_4/V_4 \cong S_3$ and $S_4 = V_4 \rtimes S_3$.
- *Next:* Lecture 3 asks what the order of a group forces about its subgroups: Sylow's theorems.

References

Dummit and Foote, *Abstract Algebra*: normal subgroups and quotients, §3.1, Prop. 5, p. 81, Thm 6 and Prop. 7, p. 82; index 2, §3.2, p. 91; the first and fourth (correspondence) isomorphism theorems, Thm 16, p. 97, and Thm 20, p. 99; sign and A_n, §3.5, pp. 107–109; classes of S_n, §4.3, Props. 10–11, pp. 125–126; direct and semidirect products, §5.4, Thm 9, p. 171, §5.5, Thm 12, p. 180. Rotman, *An Introduction to the Theory of Groups*: cycles and sign, Thms 1.1–1.6, pp. 6–9; the correspondence theorem, Thm 2.28, p. 38; conjugacy in S_n, Thm 3.5, p. 47; semidirect products, pp. 167–170. Sagan, *The Symmetric Group*: z_λ and the class size, Prop. 1.1.1 and (1.2), p. 3.